Paper #26b: Sterile Neutrino Mass Generation — UQFF (Redirect Notice)

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__Session:__ Phase 1 (Sessions 1–43)

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Author: Daniel T. Murphy **Date:** March 6, 2026 **Session:** Phase 1 (Sessions 1–43) **Framework:** UQFF Star-Magic (κ = 0.0005/day, [SSq] = 0.57, βi = *0.61) __Source:__* `CondensedPhysics2.py`, `MAIN_1_CoAnQi.cpp` ***Cross-links:*** *PAPER*026*Sterile*Neutrino*Mass*Generation*UQFF.md (canonical), PAPER*027 (Lepton Flavor Violation), PAPER_028 (BSM Coupling Constants)

Abstract

This file is a redirect stub. All sterile neutrino mass generation content — including the GUT-scale sector, Yukawa coupling matrix, seesaw mechanism, neutrinoless double beta decay constraints, and UQFF vacuum-mediated mass generation equations — has been merged into the canonical whitepaper PAPER*026*Sterile*Neutrino*Mass*Generation*UQFF.md. See that file for full derivations and numerical results.

Key parameters consolidated: sterile neutrino mass $M_N \sim$ *1e14 GeV, active-sterile mixing* $|U_{eN}|^2 < 1\text{e-}3$, vacuum contribution $\delta m_\nu = \kappa \times [SSq] \times m_N$.

MERGED INTO: *PAPER026SterileNeutrinoMassGenerationUQFF.md*

Key Equations (Summary)

The UQFF-modified seesaw mass formula:

```
m_\nu = m_D^2 / M_N + \delta m_{UQFF}
```

```
\delta m_{UQFF} = \kappa \times [SSq] \times \frac{v^2}{M_N} = 5.0 \times 10^{-4} \times
0.57 \times \frac{v^2}{M_N}
```

```
|U_{eN}|^2 < 1.0\times10^{-3} \quad \text{(experimental constraint)}
```

Numerical summary: M_N = *1.0e14 GeV,* m_D = 1.74e2 GeV (top-quark scale), $\delta m_{UQFF} \approx$ *2.85e-4* $\times$ v^2/M_N, κ[SSq] = 2.85e-4

Date merged: 2026-03-06 **Merged by:** GitHub Copilot

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0■10?4 day?■, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments. LHC direct-search results and arXiv constraints bound the parameter space.